Jinyuan Mao

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Education

Zhejiang University Hangzhou, China
B.S. in Statistics & B.Mgt. in Info. Management and Info. Systems (Dual Degree) *Sep. 2023 – Jun. 2027*

- **GPA:** 92.5/100, **Rank: 1/20**
- **Research Interests:** Generative Models, Efficient AI, Computer Vision

Publications

* denotes equal contribution

[1] **BIFE: Better Interaction, Fewer Errors for Minute-Long Video Generation**
Zeyu Zhang*, **Jinyuan Mao***, Shuning Chang, Yuanyu He, Yizeng Han, Jiasheng Tang, Fan Wang, Bohan Zhuang
SIGGRAPH Asia 2026 Technical Communications | arXiv

[2] **WorldAttention: An Efficient Attention Architecture for Interactive Video World Models**
Zeyu Zhang*, **Jinyuan Mao***, Dakai An*, Wangbo Zhao, Hanfeng Lu, Jiasheng Tang, Yinghao Yu, Wei Wang, Bohan Zhuang
NeurIPS 2026 Reviewed (5/5/5 – Accept/Accept/Accept) | arXiv

[3] **HimmBO: High-Dimensional Interactive Model-Merging Bayesian Optimization for Image Generation**
Jinyuan Mao, Zhihao Shen, Chenxi Liu
Under review at Eurographics 2027

[4] **IntuiTF: MLLM-Guided Transfer Function Optimization for Direct Volume Rendering**
Yiyao Wang, Bo Pan, Ke Wang, Han Liu, **Jinyuan Mao**, Yuxin Liu, Minfeng Zhu, Xiuqi Huang, Weifeng Chen, Bo Zhang, Wei Chen
arXiv preprint, 2025 | arXiv

Research Experience

Research Intern, PixAI Jun. 2026 – Oct. 2026
Interactive image generation; advised by Chenxi Liu (University of Victoria) *Hangzhou, China*

- As first author and sole student researcher, proposed **HimmBO** [3], a two-stage preferential Bayesian optimization framework that scales interactive adapter (LoRA) merging from a few dozen to **1080 adapters**, with a level-set screening stage that needs no image generation during interaction.
- Designed a style-aware kernel that transfers preference information across adapters; reached a given similarity with **under 1/10** of the inference required by prior work, and stayed stable as the library grew while baselines degraded.

Research Intern, ZIP Lab, Zhejiang University Oct. 2025 – Jun. 2026
Efficient video generation and world models; advised by Prof. Bohan Zhuang *Hangzhou, China*

- **WorldAttention** [2]: found that attention heads in video diffusion transformers have highly heterogeneous sparsity, and designed **Hybrid Sparse Attention**, pairing a linear global branch with head-adaptive block-sparse attention; implemented it end to end, achieving a **2.21**× inference speedup at matched quality.
- **BIFE** [1]: co-designed a semantic sparse KV cache that retrieves relevant history instead of keeping a sliding window, preserving long-range interactions in minute-long video generation; improved VDE-Subject by **22.2%** and VDE-Clarity by **19.4%**.

Project Leader, Provincial SRTP Project Apr. 2024 – Jun. 2025
Database Agent Framework *Zhejiang University*

- Proposed a "Two-Stage Trial-and-Error Learning" framework to improve database agents' interaction logic; filed one national invention patent.

Member, ZJU Supercomputing Team 2024 – Present
Student cluster competitions and HPC application optimization *Zhejiang University*

- **SC25 IndySCC** (1st place globally): built, parallelized, and tuned the SST architecture simulator, and searched for the best-performing simulation configurations.
- **ICT Competition** (1st Place, National Finals): ported a multimodal LLM (InternVL2.5) to an NPU-based distributed training stack, achieving a **1.15**× training speedup.
- **PAC 2025** (National 2nd Prize): sped up floating-point matrix kernels **4**× with multi-threading, SME vectorization, and block prefetching.

Honors & Awards

National Scholarship (Top 0.2%), three times — 2024, 2025, 2026
1st Place Globally, SC25 IndySCC Student Cluster Competition — Nov. 2025
National 2nd Prize, 12th National Parallel Application Challenge (PAC) — Aug. 2025
2nd Prize, ASC25 Student Supercomputer Challenge — May 2025
1st Place, National Finals, ICT Competition — Mar. 2025

Other Experience

AI Engineering Intern, Nodesk AI — Jun. 2025 – Aug. 2025
Algorithm development — *Hangzhou, China*

- Built a smart-reply cross-validation module, improving accuracy by **15%** via fine-tuning; co-authored a patent on a guardrail mechanism combining reranking models with scenario recognition.

Operations & Infrastructure Lead, ZJU X-Lab — Jun. 2025 – Present
Student-run innovation organization — *Zhejiang University*

- Manage the GPU clusters with Docker-isolated environments for 10+ projects; built a private Overleaf, NAS storage, VPN, and the organization's website.

Skills

Languages: Python, C/C++, SQL, LaTeX
ML & Systems: PyTorch, Diffusers, MPI, OpenMP, Docker, Linux
HPC & Profiling: CUDA, Slurm, Nsight Systems, Intel VTune, perf